#• $\frac{\partial}{\partial t}\int_{CV}\rho\,d\forall + \int_{CS}\rho\vec{v}\cdot d\vec{A} = 0 \Rightarrow \int_{CV}\frac{\partial\rho}{\partial t}d\forall + \int_{CV}\nabla\cdot(\rho v)\,d\forall = 0 \Rightarrow \partial\rho/\partial t + \nabla\cdot(\rho v) = 0$ (DIRECTLY FROM INTEGRAL FORM)

CYLINDRICAL COORDINATES → $\partial\rho/\partial t + 1/r\,\partial/\partial r(r\rho v_r) + 1/r\,\partial/\partial\theta(\rho v_\theta) + \partial/\partial z(\rho v_z) = 0$

• $\rho|_{x+dx/2} = \rho + \partial\rho/\partial x\cdot dx/2 + \frac{\partial\rho}{\partial x}\cdot\frac{(dx)^2}{2!} + \dots$ ⇒ $\int_{CV}\frac{\partial\rho}{\partial t}\,d\forall + \int_{CS}\left\{\rho + \rho\frac{dx}{2} + \rho''\frac{dx^2}{2!} + \dots\right\}\left\{u + u'\frac{dx}{2} + u''\frac{dx^2}{2!} + \dots\right\}x\,dy\,dz$

$u|_{x+dx/2} = u + \partial u/\partial x\cdot dx/2 + \partial^2 u/\partial x^2\cdot\frac{(dx)^2}{2!} + \dots$ $+\int_{CS}\left\{\rho - \rho'\frac{dx}{2} + \rho''\frac{dx^2}{2!} + \dots\right\}\left\{u - u'\frac{dx}{2} + u''\frac{dx^2}{2!} - \dots\right\}x\,dy\,dz$

+ 2 OTHER CONTROL SURFACES $= 0$

• SPECIAL CASES: (A) STEADY FLOW ⇒ $\partial/\partial t(\cdot) = 0 \Rightarrow \nabla\cdot(\rho v) = 0 \Rightarrow v\cdot\nabla\rho + \rho\nabla\cdot v = 0$

(B) INCOMPRESSIBLE FLOW ⇒ $\nabla\cdot v = 0 \Rightarrow \partial\rho/\partial t + v\cdot\nabla\rho + \rho\nabla\cdot v = 0 \Rightarrow \partial\rho/\partial t + v\cdot\nabla\rho = 0 \Rightarrow D\rho/Dt = 0$ (MATERIAL DERIVATIVE IS ZERO)

(C) ρ = CONSTANT ⇒ $\partial\rho/\partial t = 0$, $\nabla\rho = 0 \Rightarrow \nabla\cdot v = 0$ (OBVIOUSLY AS FLOW IS INCOMPRESSIBLE)

• FOR STREAMFUNCTION, $\nabla\cdot v = 0 \Rightarrow u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$, IF $\nabla\cdot(\rho v) = 0$ (STEADY FLOW FROM CONSERVATION OF MASS)
$\Rightarrow \rho u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$ (VALID WORKING MODEL)

EXAMPLE: FOR A FLOW IN THE xy PLANE, THE x-COMPONENT OF VELOCITY IS GIVEN BY $u = Ax^2y^2$, WHERE $A = 0.3\ m^{-3}s^{-1}$, AND x & y ARE MEASURED IN METERS. FIND A POSSIBLE y-COMPONENT FOR STEADY, INCOMPRESSIBLE FLOW. IS THIS ALSO VALID FOR UNSTEADY, INCOMPRESSIBLE FLOW? WHY? HOW MANY POSSIBLE y-COMPONENTS ARE THERE? DETERMINE EQUATION OF STREAMLINE FOR THE SIMPLEST y-COMPONENT OF VELOCITY. PLOT THE STREAMLINES THROUGH (1,4) AND (2,4).

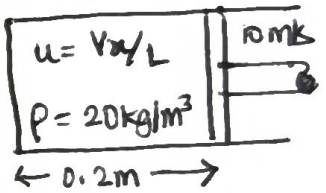
WHEN FLOW IS INCOMPRESSIBLE & 2D, WE CAN DEFINE STREAMFUNCTION IRRESPECTIVE OF WHETHER FLOW IS STEADY

$u = \partial\psi/\partial y = Ax^2y^2 \Rightarrow \psi = Ax^2y^3/3 + F(x) \Rightarrow v = -\partial\psi/\partial x = -\frac{2}{3}Axy^3 + \partial F/\partial x$ ⇒ INFINITELY MANY POSSIBLE y-COMPONENTS !!

SIMPLEST IS FOR $F(x) = 0 \Rightarrow \psi = Ax^2y^3/3 = $ CONSTANT. $\Rightarrow x^2y^3 = $ CONSTANT (EQN. OF STREAMLINE)

THROUGH (1,4) $\Rightarrow 1^2\times4^3 \Rightarrow x^2y^3 = 64$ (EQN.), THROUGH (2,4) $\Rightarrow 2^2\times4^3 \Rightarrow x^2y^3 = 256$ (EQN.)

EXAMPLE: A GAS-FILLED PNEUMATIC STRUT IN AN AUTOMOBILE SUSPENSION SYSTEM BEHAVES LIKE A PISTON–CYLINDER APPARATUS. AT ONE INSTANT WHEN THE PISTON IS 0.2m AWAY FROM THE CLOSED END OF THE CYLINDER, $\rho = 20\ kg/m^3$ AND PISTON BEGINS TO MOVE AWAY @ 10m/s. ASSUME AS A SIMPLE MODEL THAT GAS VELOCITY IS 1D & PROPORTIONAL TO DISTANCE FROM CLOSED END; VARYING LINEARLY FROM 0 TO V. WHAT IS RATE OF CHANGE OF GAS DENSITY AT THIS INSTANT. OBTAIN AN EXPRESSION FOR AVERAGE DENSITY VERSUS TIME.



ASSUME GAS DENSITY IS UNIFORM IN SPACE ⇒ $\partial\rho/\partial t + \rho\,\partial u/\partial x = 0$ (NO OTHER VELOCITY COMPONENTS)

⇒ RATE OF CHANGE $= -\frac{20\times10}{0.2} = -1000\ \frac{kg}{m^3s}$

$\Rightarrow \frac{\partial\rho}{\partial t} + \rho\cdot V/L = 0 \Rightarrow \int\frac{d\rho}{\rho} = \int\frac{-V}{L}\cdot dt \Rightarrow \ln(\rho/\rho_0) = -V/L(t - t_0)$ IS THIS CORRECT? L ALSO VARIES WITH t??

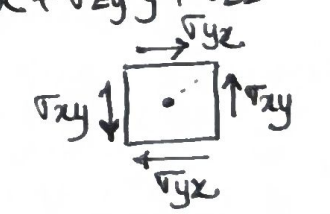
$\Rightarrow \int\frac{d\rho}{\rho} = \int\frac{-V}{L_0 + Vt}\cdot dt \Rightarrow \rho(t) = \rho_0\left\{\frac{1}{1 + Vt/L_0}\right\}$ → FINAL EXPRESSION.

• $\rho\,\delta h\cdot\delta A\cdot Dv/Dt = T^{(n)}\cdot\delta A + T^{(-n)}\cdot\delta A + \rho\,\delta h\,\delta A\cdot G$ (FORCE BALANCE) $\Rightarrow \lim_{\delta h\to0} T^{(n)} = -T^{(-n)}$ } EQUIVALENT TO NEWTON'S 3RD LAW OF ACTION & REACTION.

• GEOMETRY, $\delta A_x = \delta A n_x$, $\delta A_y = \delta A n_y$, $\delta A_z = \delta A n_z \Rightarrow \rho\,\forall\,Dv/Dt = \rho\,\forall\,\vec{G} + T\cdot\delta A + T^{(-x)}\cdot\delta A_x + T^{(-y)}\delta A_y + T^{(-z)}\cdot\delta A_z$

$\Rightarrow \lim_{V\to0} T = T^{(x)}\cdot n_x + T^{(y)}n_y + T^{(z)}\cdot n_z$ } COMPONENTS OF TRACTION ARE NECESSARY TO CREATE STRESS TENSOR

$T^{(x)} = \sigma_{xx}\hat{x} + \sigma_{xy}\hat{y} + \sigma_{xz}\hat{z}$, $T^{(y)} = \sigma_{yx}\hat{x} + \sigma_{yy}\hat{y} + \sigma_{yz}\hat{z}$, $T^{(z)} = \sigma_{zx}\hat{x} + \sigma_{zy}\hat{y} + \sigma_{zz}\hat{z}$



• $I_z\frac{d\dot{\theta}_z}{dt} = \text{TORQUE}_z$, $I_z = \int_{\delta z/2}^{-\delta z/2}\int_{\delta y/2}^{-\delta y/2}\int_{\delta x/2}^{-\delta x/2}(x^2+y^2)\rho\,dx\,dy\,dz = \rho\,\delta x\,\delta y\,\delta z\left\{\frac{\delta x^2+\delta y^2}{12}\right\}$

(BARRING TORQUE GENERATING BODY FORCES)

$\text{TORQUE}_z = 2\cdot\delta x/2\cdot\sigma_{xy}\cdot\delta y\,\delta z - 2\,\delta y/2\,\sigma_{yx}\cdot\delta y\,\delta z = (\sigma_{xy} - \sigma_{yz})\,\delta x\,\delta y\,\delta z$

$\Rightarrow \rho\,\delta x\,\delta y\,\delta z\left\{\frac{\delta x^2+\delta y^2}{12}\right\}\cdot\frac{d\dot{\theta}_z}{dt} = (\sigma_{xy} - \sigma_{yx})\,\delta x\,\delta y\,\delta z \Rightarrow \lim_{\delta x,\delta y\to0}\sigma_{xy} = \sigma_{yz}$ } CONSERVATION OF ANGULAR MOMENTUM.

WHATEVER IS DISCUSSED SO FAR IS GENERAL CONTINUUM MECHANICS.

• NEWTONIAN MODEL: $D = \frac{1}{2}(\nabla u + \nabla u^T)$, $\sigma = -p\mathbb{I} + \lambda\,\text{TRACE}(D) + 2\mu D$ → ISOTROPY, LINEARITY.

μ = QUANTIFIES FLUID RESISTANCE TO SHEARING.

λ = ADDITIONAL RESISTANCE TO VOLUMETRIC EXPANSION/CONTRACTION.

p = THERMODYNAMIC PRESSURE.

$\Rightarrow \text{TRACE}(\sigma) = -3p + 3\lambda\nabla\cdot v + 2\mu\nabla\cdot v \Rightarrow \underbrace{-\frac{1}{3}\text{TRACE}(\sigma)}_{\text{MECHANICAL PRESSURE}} = \underbrace{p}_{\text{THERMODYNAMIC PRESSURE}} - \nabla\cdot v\,\underbrace{(\lambda + 2\mu/3)}_{\text{STOKES' HYPOTHESIS}}$

• FINALLY, $\sigma = -p\mathbb{I} + \mu\left\{\nabla v + \nabla v^T - \frac{2}{3}\nabla\cdot v\,\mathbb{I}\right\}$ → NEWTONIAN FLUID MODEL.

• TAYLOR SERIES: $\left(\tau_{xx}+\frac{\partial \tau_{xx}}{\partial x}\frac{dx}{2}\right)dydz-\left(\tau_{xx}-\frac{\partial \tau_{xx}}{\partial x}.\frac{dx}{2}\right)dydz=\frac{\partial \tau_{xx}}{\partial x}.dxdydz$: NORMAL COMPONENT OF STRESS ALONG x.

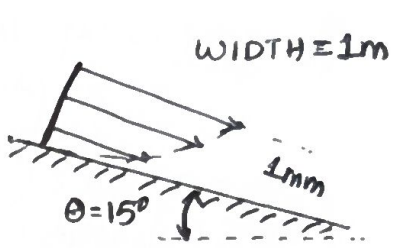
• GENERALIZED CAUCHY STRESS EQUATION: $\rho\, Dv/Dt=\nabla.\tau+\rho b$; v = EULERIAN VELOCITY, τ = CAUCHY STRESS TENSOR, b = GENERALIZED BODY FORCE PER UNIT MASS.

• NEWTONIAN FLUID: $\tau=-pI+\mu\{\nabla v+\nabla v^T-\frac{2}{3}\nabla.v\,I\}$

$\tau_{xx}=-p+\mu\{2\partial u/\partial x-\frac{2}{3}(\partial u/\partial x+\partial v/\partial y+\partial w/\partial z)\}$, $\tau_{yy}=-p+\mu\{2\partial v/\partial y-\frac{2}{3}(\partial u/\partial x+\partial v/\partial y+\partial w/\partial z)\}$

$\tau_{xy}=\tau_{yx}=\mu\{\partial u/\partial y+\partial v/\partial x\}$, $\tau_{xz}=\tau_{zx}=\mu\{\partial u/\partial z+\partial w/\partial x\}$, $\tau_{yz}=\tau_{zy}=\mu\{\partial v/\partial z+\partial w/\partial y\}$

EXAMPLE: A LIQUID FLOWS DOWN AN INCLINED PLANE SURFACE IN A STEADY, FULLY DEVELOPED FILM OF THICKNESS h. SIMPLIFY THE CONTINUITY & NS EQNS. TO MODEL THIS FLOW FIELD. OBTAIN EXPRESSIONS FOR THE LIQUID VELOCITY PROFILE, SHEAR STRESS DISTRIBUTION, VOLUME FLOW RATE & AVERAGE VELOCITY. RELATE THE LIQUID FILM THICKNESS TO THE VOLUME FLOW RATE PER UNIT DEPTH OF SURFACE NORMAL TO THE FLOW. CALCULATE THE VOLUME FLOW RATE IN A FILM OF WATER 1 mm, FLOWING ON A SURFACE 1m WIDE, INCLINED AT 15° TO THE HORIZONTAL.



CONSERVATION OF MASS / CONTINUITY ≡ $\partial u/\partial x+\partial v/\partial y+\partial w/\partial z=0 \Rightarrow \partial v/\partial y=0$ (0 (FULLY DEVELOPED); ∂/∂z = 0)

x-MOMENTUM ≡ $\rho(\partial u/\partial t+u\partial u/\partial x+v\partial u/\partial y+w\partial u/\partial z)=\rho b_x-\partial p/\partial x+\mu(\partial^2u/\partial x^2+\partial^2u/\partial y^2+\partial^2u/\partial z^2)$

STEADY; DEVELOPED; w = 0; DEVELOPED; DEVELOPED; ∂/∂z(.) = 0

Y-MOMENTUM ≡ $\rho(\partial v/\partial t+u\partial v/\partial x+v\partial v/\partial y+w\partial v/\partial z)=\rho b_y-\partial p/\partial y+\mu(\partial^2v/\partial x^2+\partial^2v/\partial y^2+\partial^2v/\partial z^2)$

STEADY; DEVELOPED; COM; w = 0; DEVELOPED; COM; ∂/∂z(.) = 0

FROM DEVELOPED $\partial v/\partial x=0$, FROM COM $\partial v/\partial y=0$ & FROM NOTHING VARIES $\partial v/\partial z=0 \Rightarrow$ V = CONSTANT.

HOWEVER, @ TOP BOUNDARY & BOTTOM WALL V = 0 ⇒ CONSTANT = 0

∴ x-MOMENTUM ≡ $\mu\frac{\partial^2u}{\partial y^2}+\rho g\sin\theta=0$, Y-MOMENTUM ≡ $-\frac{\partial p}{\partial y}-\rho g\cos\theta=0$

WE ALSO KNOW THAT @ $y=0$, $u=0$ & @ $y=h$, $\partial u/\partial y=0 \Rightarrow u=\rho g\sin\theta/\mu\{hy-y^2/2\}$.

$\tau_{xy}=\tau_{yx}=\mu\{\frac{\partial u}{\partial y}\}=\rho g\sin\theta\{h-y\}$, $Q=\int u.dA=\int\frac{\rho g\sin\theta}{\mu}(hy-y^2/2).bdy=\frac{\rho g\sin\theta}{\mu}.\frac{bh^3}{3}$

$\Rightarrow V_{AVERAGE}=\frac{Q}{bh}=\frac{\rho g\sin\theta}{\mu}.\frac{h^2}{3}$.

• SINCE NO NATURAL OR FUNDAMENTAL UNITS OF MEASUREMENT EXIST FOR A PHYSICAL VARIABLE, WE CAN TRANSFORM GOVERNING EQUATIONS INTO MORE AMENABLE FORMS USING SCALES FROM OUR PROBLEM.

• $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$, $\rho\{u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}\}=-\frac{\partial p}{\partial x}+\mu\{\frac{\partial^2u}{\partial x^2}+\frac{\partial^2u}{\partial y^2}\}$, $\rho\{u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}\}=-\frac{\partial p}{\partial y}+\mu\{\frac{\partial^2v}{\partial x^2}+\frac{\partial^2v}{\partial y^2}\}-\rho g$

SCALING: $x=L\bar{x}$, $y=L\bar{y}$, $u=V\bar{u}$, $v=V\bar{v}$, $p=P\bar{p}$

$\Rightarrow \frac{\partial\bar{u}}{\partial\bar{x}}+\frac{\partial\bar{v}}{\partial\bar{y}}=0$, $\bar{u}\frac{\partial\bar{u}}{\partial\bar{x}}+\bar{v}\frac{\partial\bar{u}}{\partial\bar{y}}=-\frac{P}{\rho V^2}\frac{\partial\bar{p}}{\partial\bar{x}}+\frac{\mu}{\rho VL}\{\frac{\partial^2\bar{u}}{\partial\bar{x}^2}+\frac{\partial^2\bar{u}}{\partial\bar{y}^2}\}$, $\bar{u}\frac{\partial\bar{v}}{\partial\bar{x}}+\bar{v}\frac{\partial\bar{v}}{\partial\bar{y}}=-\frac{P}{\rho V^2}\frac{\partial\bar{p}}{\partial\bar{y}}-\frac{gL}{V^2}+\frac{\mu}{\rho VL}\{\frac{\partial^2\bar{v}}{\partial\bar{x}^2}+\frac{\partial^2\bar{v}}{\partial\bar{y}^2}$

(EQUIVALENT SYSTEM OF EQUATIONS, BUT WITH SIGNIFICANT REDUCTION IN VARIABLES).

• $g(q_1,q_2,\ldots,q_n)=0 \Rightarrow G(\pi_1,\pi_2,\ldots,\pi_{n-m})=0$ (BUCKINGHAM-PI THEOREM) → PI PARAMS ARE INDEPEN

• CARE SHOULD BE TAKEN TO ENSURE THAT THE PI GROUPS ARE INDEPENDENT.

EXAMPLE: THE PRESSURE DROP, Δp, FOR STEADY, INCOMPRESSIBLE VISCOUS FLOW THROUGH A STRAIGHT HORIZONTAL PIPE DEPENDS ON PIPE LENGTH ℓ, AVERAGE VELOCITY V, FLUID VISCOSITY μ, PIPE DIAMETER D, FLUID DENSITY ρ, AND THE AVERAGE ROUGHNESS HEIGHT e. DETERMINE A SET OF DIMENSIONLESS GROUPS THAT CAN BE USED TO CORRELATE DATA.

FROM GIVEN ANSATZ, $g(\Delta p,\ell,V,\mu,D,\rho,e)=0 \Rightarrow n=7$

NOW, $[\Delta p]=ML^{-1}T^{-2}$, $[\ell]=L$, $[V]=LT^{-1}$, $[\mu]=ML^{-1}T^{-1}$, $[D]=L$, $[\rho]=ML^{-3}$, $[e]=L \Rightarrow m=3$

⇒ WE NEED TO CHOOSE 3 INDEPENDENT PARAMETERS, μ, V, ρ (THIS CAN ENCOMPASS EVERYTHING)

REMAINING, $[\Delta p]\mu^bV^c\rho^d=[0] \Rightarrow (ML^{-1}T^{-2})(ML^{-1}T^{-1})^b(LT^{-1})^c(ML^{-3})^d=[0]$

$\Rightarrow M^{1+b+d}L^{-1-b+c-3d}T^{-2-b-c}=[0]$

$\Rightarrow b+d=-1$, $-b+c-3d=1$, $b+c=-2 \Rightarrow 2c-3d=-1$, $2b+3d=-3 \Rightarrow d=-1$

$\Rightarrow b=0, c=-2, d=-1 \Rightarrow \Delta p.V^{-2}\rho^{-1} \Rightarrow \pi_1=\Delta p/\rho V^2$

SIMILARLY, $[\ell]\mu^bV^c\rho^d=[0] \Rightarrow (L)(ML^{-1}T^{-1})^b(LT^{-1})^c(ML^{-3})^d=[0]$

$\Rightarrow M^{b+d}L^{1-b+c-3d}T^{-b-c}=[0] \Rightarrow b+d=0, 1-b+c-3d=0, b+c=0$

$\Rightarrow 2b-1+3d=0, b+d=0 \Rightarrow d=1, b=-1, c=1 \Rightarrow \pi_2=\rho\ell V/\mu$

SIMILARLY, $\pi_3=\rho\ell D/\mu$, $\pi_4=\rho\ell e/\mu$

$\Rightarrow G\left(\frac{\Delta p}{\rho V^2},\frac{\rho\ell V}{\mu},\frac{\rho\ell D}{\mu},\frac{\rho\ell e}{\mu}\right)=0$ } SET OF DIMENSIONLESS GROUPS THAT CAN BE USED TO CORRELATE DATA.